Controllable Idle Motion Generation with Flow Matching

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Motivation

- **Idle animations** are **essential for character believability**
- Games and interactive applications use **prerecorded** idle **animations**
- These are **costly**, and **not flexible nor controllable** in-game
- Deep learning and **generative systems** provide:
  - **Conditioning** over the output (controllability)
  - **Stochastic** generation (variability)
- The generated animations could be potentially used as a **starting canvas** for further refining by artists or **adapted as an in-game system**

Contributions

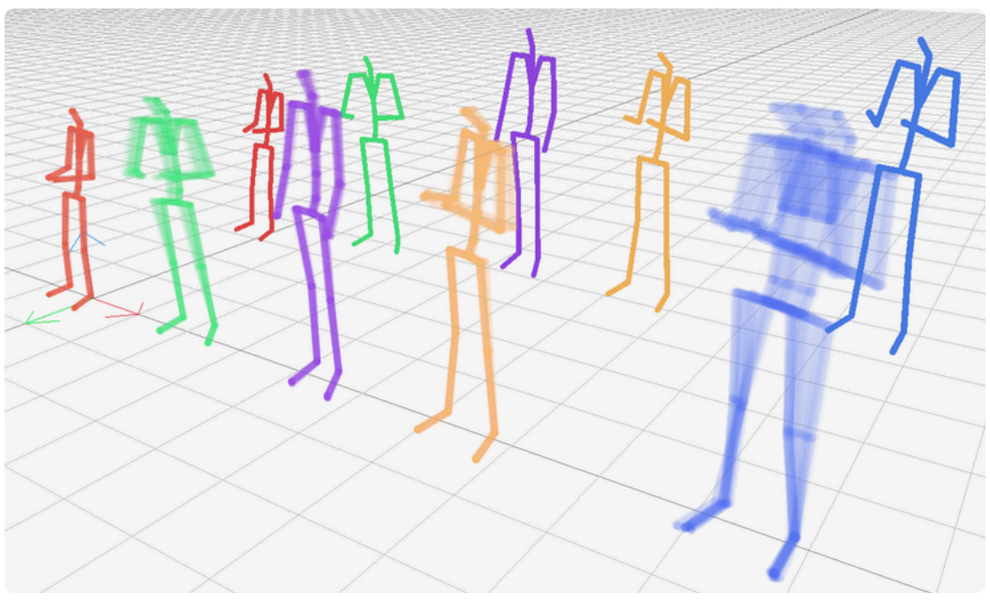
- A flow matching **network** that **generates idle animations** *(achieved)*
- A **conditioning system** based on pose and joint speed to **control the style** of the generated motion *(achieved)*
- An **analysis** of diverse **control systems** *(work in progress)*

Note: this is a work in progress, we plan on further expanding the research.

Control over key output characteristics

Average pose

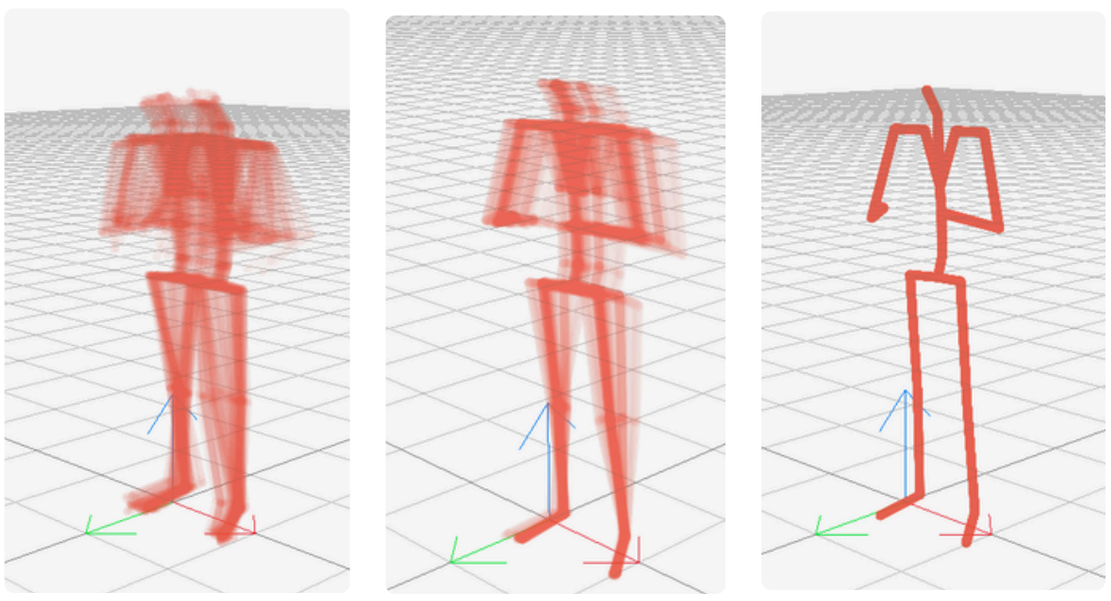
How does the person stand? Do they have their arms on their hips? Do they cross their arms? Are they using a phone? How do they position their legs?



Different input average poses (the row on the back) generate different animations that adhere to these poses

Joint speeds

How does the person move? Do they stand very still? Do they move their hands a lot?



Speed input: fastest | Speed input: average | Speed input: slowest

Details

Network

- **Transformer U-net**, based on Match-TTSG (Mehta et al.)

Data

- **StayStill**, a large-scale 3D idle animation dataset
- Automatic filtering and **removal of high-speed parts**
- **Data augmentation**: mirroring, root randomisation

Technique

- **Conditional Flow Matching**
  - **Stochastic** generation: same input, different output
  - Learns a **time-dependent velocity field** that transports a source distribution to the target data distribution.

Check the repository

The repository does **not currently contain the code**, but you can give it a **star** to follow the development of the project.

Planned release: after doing an analysis of the quality of the generated output and the quality of the control scheme. The code is planned to be released alongside the Arvix preprint.

https://github.com/Enekoassets/CtrlIdle

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